

PRITHVIRAJ CHAUDHURI

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TECHNICAL SKILLS

Programming Languages - Backend : Python, Java
Front End: React, Typescript, JQuery, Javascript.
Frameworks - LangGraph, Google ADK, Langchain, FastAPI, AEM, Apache Sling, Java Spring Boot, Django
Platforms - OpenShift, Ansible, AWS, EKS, OpenEdx, ServiceNow, Google Cloud Platform
Other Skills - Datastores: MySQL, PostgreSQL, JCR
Gitlab CI/CD pipelines, Helm Charts, Kubernetes, Linux, Bash, APIs, OpenStack

EDUCATION

Master of Computer Science North Carolina State University (NCSU), Raleigh NC	August 2019 – May 2021	GPA – 3.92
Bachelor of Technology , Computer Science and Engineering RCC Institute of Information Technology (Affiliated to Maulana Abul Kalam Azad University of Technology), Kolkata, India	August 2012 – June 2016	GPA – 8.71/10

WORK EXPERIENCE

Principal Software Applications Engineer , Red Hat	June 2021 – Present
- Developed features for an AI Agent platform to enable users to build no code agents around the langgraph deep agent harness - Built custom agents for gemini enterprise using MCP, A2A and tools defined within the Agent. - Designed RAG use cases integrating AEM, Google Drive documents with Gemini Enterprise using custom Gemini Enterprise connector to enable users to retrieve answers from AEM, using Python. - Used LangGraph and Gemini on a FastAPI service to implement an sql bot to enable internal user search mechanism on our people application, using AI to accommodate for complex queries with natural language that are too difficult to implement with standard filter searches. - Contributed to the development of platform and LLM agnostic RAG framework in collaboration with other teams to ease retrieval of internal documentation, using LangChain, Python and Chatbot built on Quarkus - Customized sales and marketing platform, AEM with React, Java and JCR to streamline asset lifecycle - Migrated applications and hosts from legacy data centers to OpenStack and Openshift VMs using OpenStack CLI with Linux OS images and ansible playbooks to modernize infrastructure and create consistent provisioning mechanisms - Implement ad hoc scripts using Python, bash etc. as and when required for various stories to run asynchronous jobs - Identified issues and developed issues with multithreaded applications and distributed transactions for faster execution times. - Design and develop CI/CD workflows to deploy application infrastructure, run automated tests using OpenShift and Gitlab	
Graduate Research Assistant , North Carolina State University	January 2020 – Present
- Implemented custom components (XBlocks) on OpenEdx for smart course design and migration of courses between instances - Integrated research projects to enable more intelligent and dynamic course generation in OpenEdx with Java, Python and Django - Built dashboards with Django and converted apps to container deployments, for consistency of deployment and maintenance	
Full Stack Web Developer Intern , Red Hat	May 2020 – August 2020
- Used Java Spring Boot and Pug/Jade to implement bulk upload of data to resource management application - Enhanced search application to enable smart searching of employee records in the organization - Designed and developed automated deployment driven by Jenkins and Git for faster delivery times and easy maintenance	
Systems Engineer , TATA Consultancy Service Ltd, Kolkata India	July 2016 – August 2019
- Implemented Service Catalog on ServiceNow for end users to utilize ServiceNow ITSM modules using CMS - Integrated ServiceNow with external systems to streamline information flow - Automated tasks by translating manual processes to system and leveraging ServiceNow orchestration platform - Designed and developed a unified REST API integration application on ServiceNow to enable faster integration - Migrated Service Catalog hosted in CMS to Service Portal using AngularJS	

PROJECTS

- Improved performance of simulation algorithms by leveraging parallel device and multithreaded capabilities of CUDA, OpenACC, OpenMP and MPI. Optimized algorithm by carefully managing memory operations
- Studied performance benefits of the same algorithm by leveraging parallel processing using CUDA, OpenACC, OpenMP and MPI
- Studied performance of various transformer networks, CNNs and word embeddings in terms of Bleu Score and Cosine similarity on flickr9k dataset